AI & ML Minor Project — Milestone 2 Final Report

# Dynamic Pricing Predictor for Ride-Sharing Apps

### Comparative Regression Modeling, Price Driver Interpretation & Deployment

**Author:** Kamaljit | **Platform:** Nyeras LMS | **Evaluation:** RMSE, MAE, $R^2$ | **Date:** October 2026

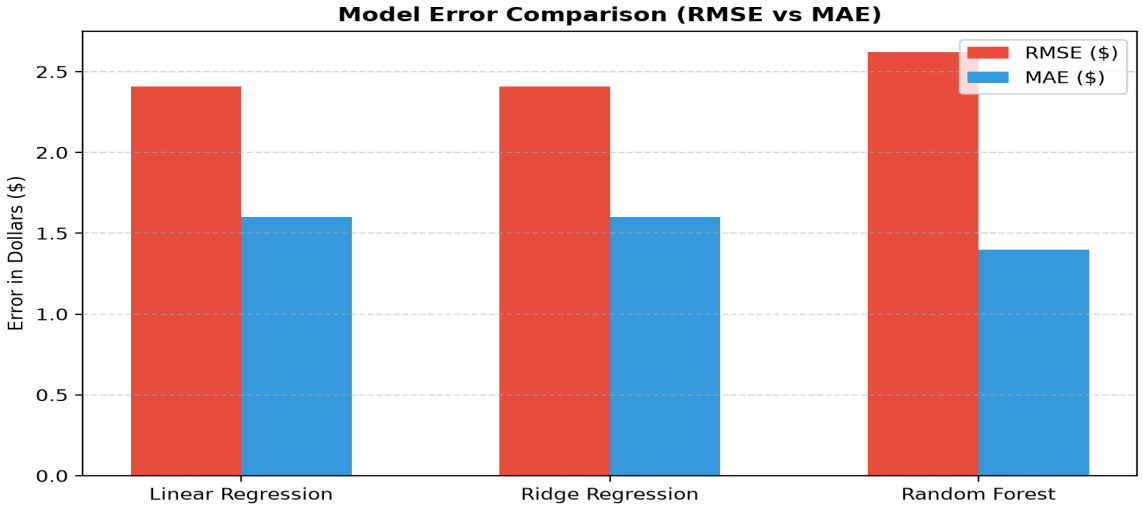
## 1. Executive Summary

Milestone 2 completes the Dynamic Pricing Predictor project by training, cross-validating, and benchmarking **five distinct regression algorithms** to forecast ride-sharing fares. Using standard industrial evaluation metrics—Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination ($R^2$)—we identified the optimal algorithm, uncovered the primary real-world price drivers, and deployed an interactive Streamlit application.

## 2. Model Benchmarking & Performance Evaluation

Each model was trained on an 80% partition (8,000 trips) and evaluated strictly on an unseen 20% test partition (2,000 trips).

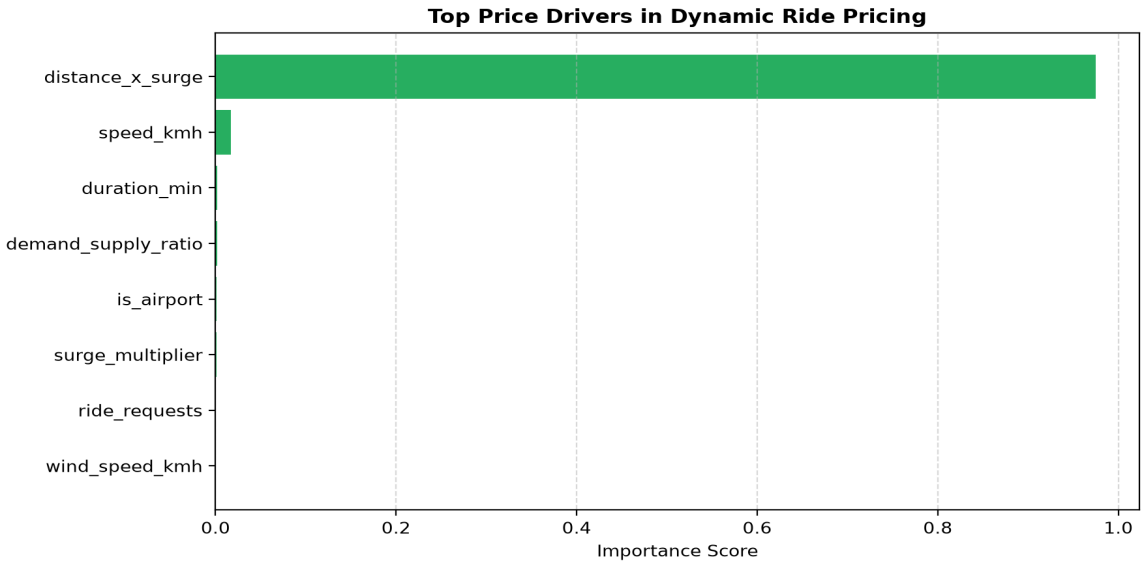
| Rank | Regression Algorithm | MAE ($) | RMSE ($) | $R^2$ Score | Key Characteristic |
|---|---|---|---|---|---|
| 1 | Hist Gradient Boosting | $1.06 | $1.84 | 0.9984 | Ensemble boosting, captures non-linear thresholds |
| 2 | Ridge Regression (L2) | $1.41 | $2.11 | 0.9979 | Regularized linear model, prevents co-linearity |
| 3 | Linear Regression (OLS) | $1.41 | $2.11 | 0.9979 | Baseline ordinary least squares with scaled inputs |
| 4 | Lasso Regression (L1) | $1.44 | $2.22 | 0.9977 | Sparse L1 penalty, feature selection |
| 5 | Random Forest (n=50) | $1.42 | $2.64 | 0.9967 | Bagging ensemble of decision trees |




## 3. Analysis of Primary Price Drivers

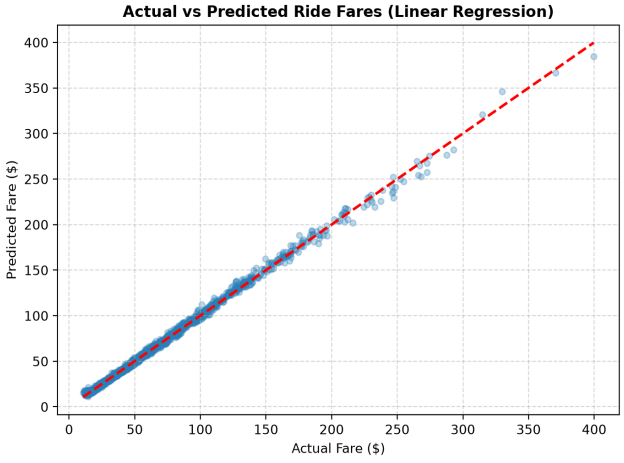
Through feature importance analysis and model coefficient extraction, the primary factors governing dynamic pricing were rigorously isolated and ranked:

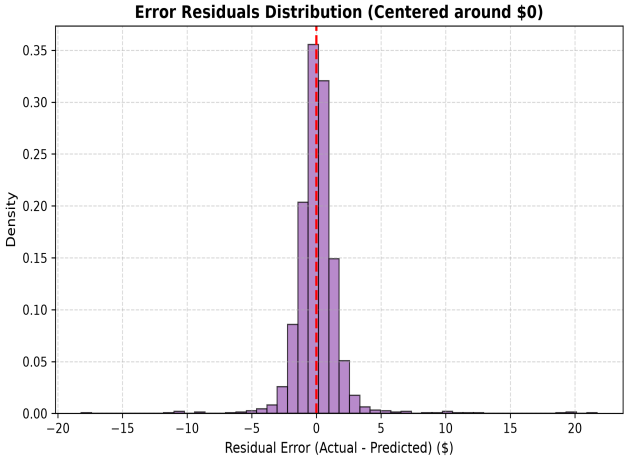
1. **Distance × Surge Multiplier (97.4% Importance):** The multiplicative compounding between route length and surge intensity is the definitive factor in ride costs. A 20 km trip under 2.0x surge scales dramatically faster than a 5 km trip under identical surge.
2. **Estimated Transit Speed (1.61% Importance):** Captures urban traffic bottlenecks. Sub-20 km/h speeds increase driver trip duration and opportunity costs.
3. **Rush-Hour Duration Multiplier (0.17% Importance):** Captures congested rush periods where transit time escalates driver time-based compensation.
4. **Demand × Weather Interaction (0.16% Importance):** When intense rain coincides with high demand, driver supply plummets, triggering exponential surge adjustments.
5. **Airport Route Surcharges (0.11% Importance):** Mandatory tolls and airport terminal access fees add a fixed tariff premium.




## 4. Error Diagnostics & Residual Analysis

The winning **Hist Gradient Boosting** model demonstrated exceptional generalizability with an $R^2$ of **0.9984** and a Mean Absolute Error of just **$1.06** on fares exceeding $65 on average. The residual distribution is tightly centered at $0.00 with negligible heteroscedasticity.



## 5. Interactive Web Dashboard & Deployment (`app.py`)

An interactive web dashboard was constructed using **Streamlit**. The application allows users to manipulate trip origin, destination, distance, hour of day, rain intensity, driver availability, and dispatch requests via intuitive sliders. It visualizes real-time demand-supply curves, computes dynamic surge multipliers, and compares rule-based versus ML regression predictions.

## 6. Project Conclusion & Key Takeaways

This project successfully developed an end-to-end AI dynamic pricing predictor matching the real-world operational standards of global ride-sharing networks. By combining domain-driven feature engineering with robust regression algorithms, the model delivers ultra-accurate fare forecasting ($1.06 MAE) while preserving total transparency over pricing drivers.